

Park, Chong Won

06. Image and Graphics Technologies



6.1 Basic Concepts of Images and Graphics
6.2 I / O Device
6.3 Processing and compression of images
6.4 2D / 3D Graphics
6.5 File formats for images and graphics

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6.1 Basic Concepts of Images and Graphics



- Images and graphics
  - Visual information
    - Visual information takes up a large part of human-accepting information
  - Image
    - Pictures generated using an input device such as a scanner or digital camera
  - Graphic
    - Pictures created through computer software such as Illustrator



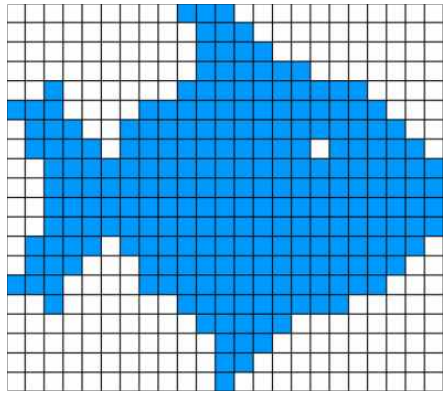
Synthesis of images and graphics

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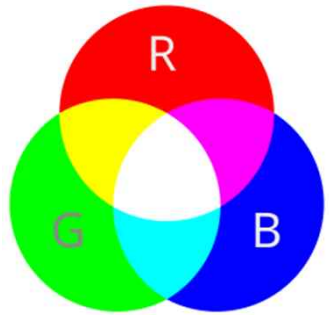
6.1 Basic Concepts of Images and Graphics



Understanding pixels



- Pixel
  - Synthesis of Picture Element : Picture + Element
  - The basic unit that constitutes the screen
  - Images are represented as a set of pixels, stored in a bitmap format
- Number of colors
  - Each pixel represents a color by appropriately combining the values of Red, Green and Blue.
  - The number of colors is proportional to bits per pixel: 2k colors
  - That is, the larger the number of allocated bits (Depth), the more color is possible



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6.1 Basic Concepts of Images and Graphics



Number of bits and color

Number of bits	The number of colors that can be represented	Remarks
1	2 (2^1)	Black & white
4	16 (2^4)	Using palettes (index color)
8	256 (2^8)	Using palettes (index color)
16	65536 (2^{16})	High Color (R:G:B = 5:5:5)
24	16,777,216 (2^{24})	True Color (R:G:B = 8:8:8)
32	16,777,216 + 8bit alpha channel	True Color + alpha channel

※ Alpha channel represents transparency

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6.1 Basic Concepts of Images and Graphics



- Number of bits and color



1 bit



4 bits



8 bits



16 bits

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6.1 Basic Concepts of Images and Graphics



Resolution

- Device resolution
  - Expressed as the number of pixels or points that can be displayed per unit length
  - Inches are often used in units of length, in which case the unit of resolution is dpi (dot per inch)
  - Printer, scanner : 300 to 700 dpi or higher / monitor : 85 to 120 dpi or higher
- Image resolution
  - Device independent image resolution
  - Represented by the total number of pixels (or horizontal and vertical pixels)
  - Applied to resolutions such as display, camera, image etc.

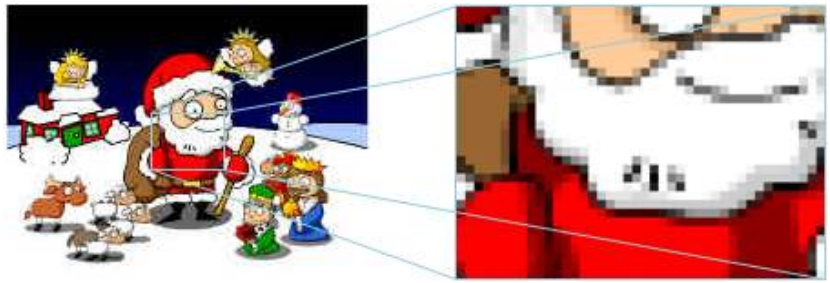
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6.1 Basic Concepts of Images and Graphics



Image and Raster / Vector Graphics

- Raster graphics
  - The method of storing in pixel unit → image, still image
  - Picture quality deteriorates when the screen is enlarged. : Staircase phenomenon
  - File size is proportional to resolution
  - Used by the painting tool and photo editing tool
- Image
  - Pixel unit as like raster graphics



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6.1 Basic Concepts of Images and Graphics



Image and Raster / Vector Graphics

- Vector graphics
  - Is represented by a geometric object (i.e. a graphical function) of points, lines, curves, and circles,
  - There is no change in image quality when enlarging the screen.
  - Generally, the file size is smaller than the raster graphics method.
  - Create geometric objects such as points / lines / circles / polygons in the drawing tool
  - Suitable for illustration, 3D graphics, animation, etc.



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6.1 Basic Concepts of Images and Graphics



Color Model

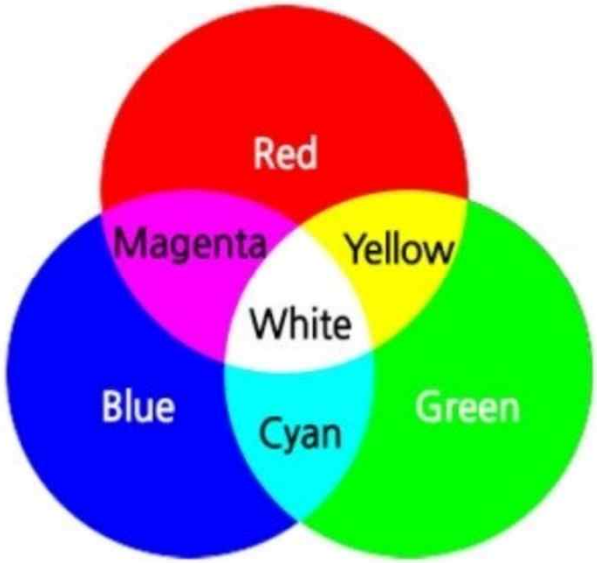
- A way to describe the characteristics of a color in certain situations
  - One color model can not be used to describe all properties of color
  - Therefore, several kinds of color models can be defined and used according to the situation.
- Types of color models
  - RGB, CMY, HSV, etc.
  - Usually, three elements are used to represent the color, so it corresponds to the three-dimensional coordinate system and each color is represented by one point

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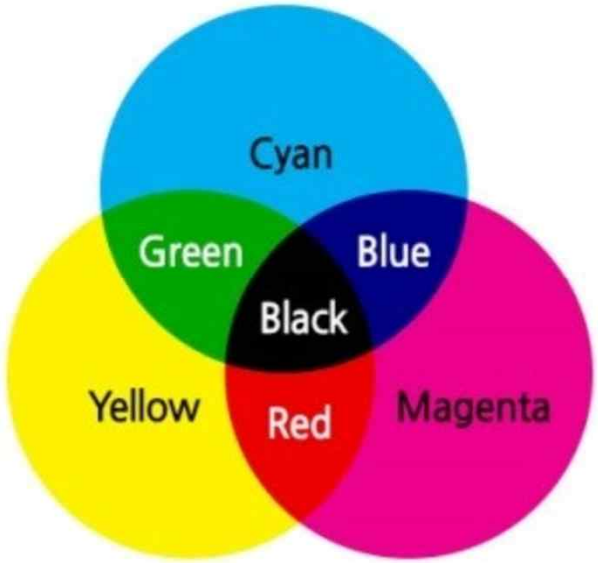
6.1 Basic Concepts of Images and Graphics



- 3 Elements of Color



Additive Colors



Subtractive Colors

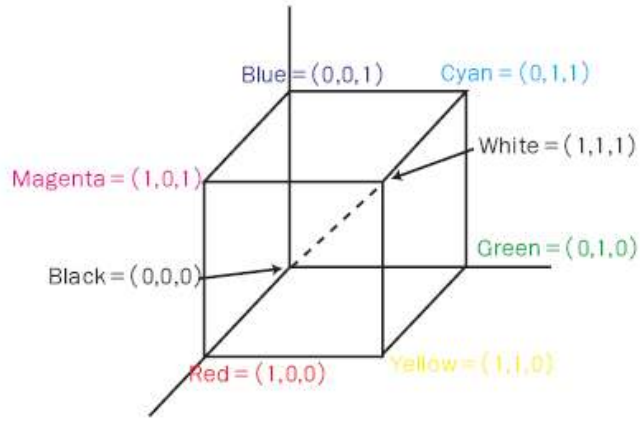
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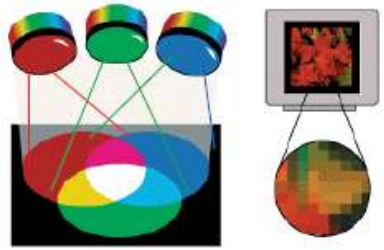
6.1 Basic Concepts of Images and Graphics



Color Model : ① **RGB (Red, Green, Blue) model**

- Additive model
  - Color model is three primary colors of light (red, green and blue) that are primary colors
  - Create a new color by adding three basic colors
  - Usage: a place to express color using light properties, a CRT monitor, etc.





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6.1 Basic Concepts of Images and Graphics



Color Model : ② **CMY (Cyan, Magenta, Yellow) model**

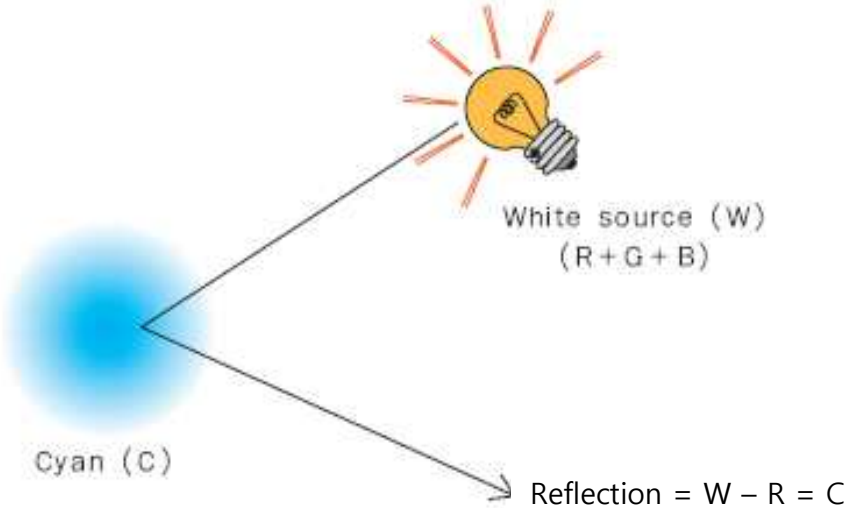
- Subtractive model
  - Color model based on secondary colors caused by light mixing
  - The three primary colors Cyan, Magenta, and Yellow are complementary colors of Red, Green, Blue,
  - Used for color printers and printing using properties such as paints and inks
  - Many use CMYK models: K (Kappa) - Black
  - Mixing CMY produces black but is not satisfactory and wastes ink

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6.1 Basic Concepts of Images and Graphics



- Color Model : ② **CMY (Cyan, Magenta, Yellow) model**



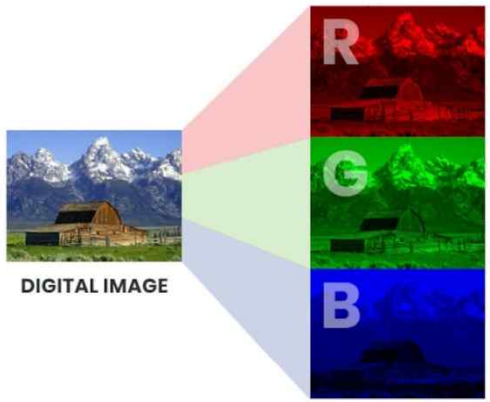
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} W \\ W \\ W \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



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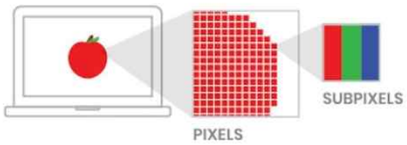
6.1 Basic Concepts of Images and Graphics





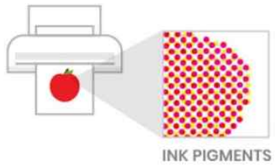


USE RGB FOR DIGITAL



e.g., web graphics or videos

USE CMYK FOR PRINT



e.g., posters or newsletters

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6.1 Basic Concepts of Images and Graphics



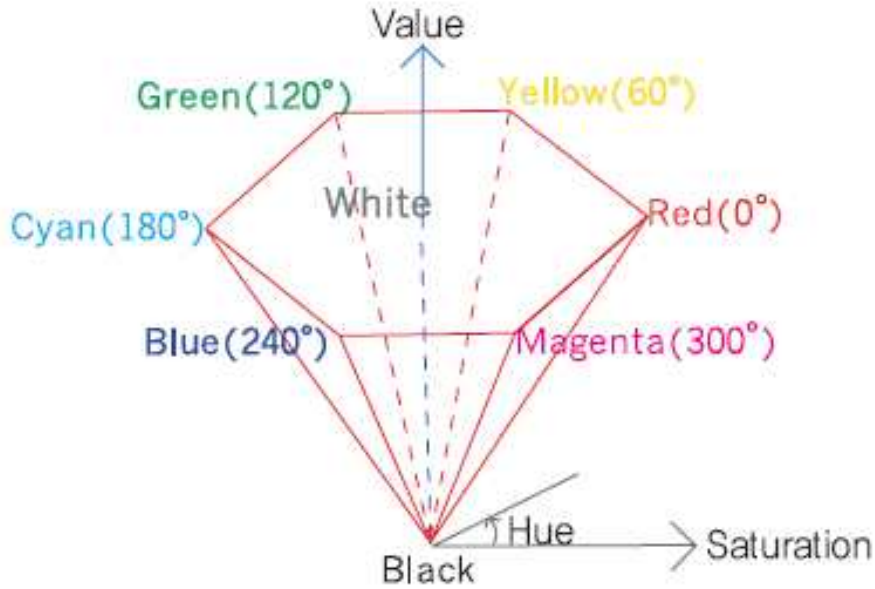
- Color Model : ③ **HSV or HSB model**
  - Similar to human intuitive visual model
    - Use three attributes: Hue, Saturation and Brightness
    - Human is distinguished from 128 (H) * 130 (S) * 23 (B) = 382,720 color
  - Convertable between RGB, CMY and HSV models

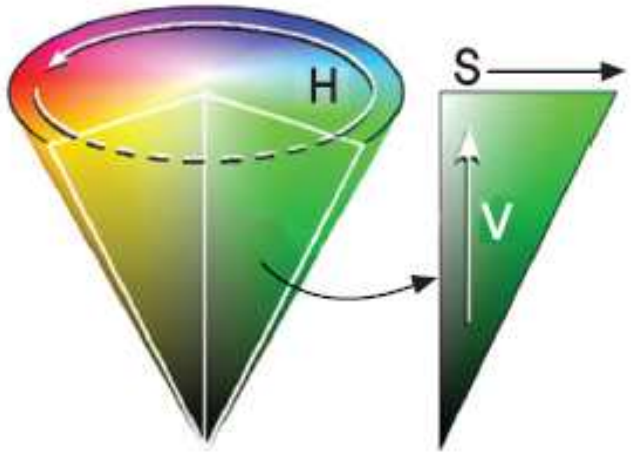
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6.1 Basic Concepts of Images and Graphics



- Color Model : ③ **HSV or HSB model**



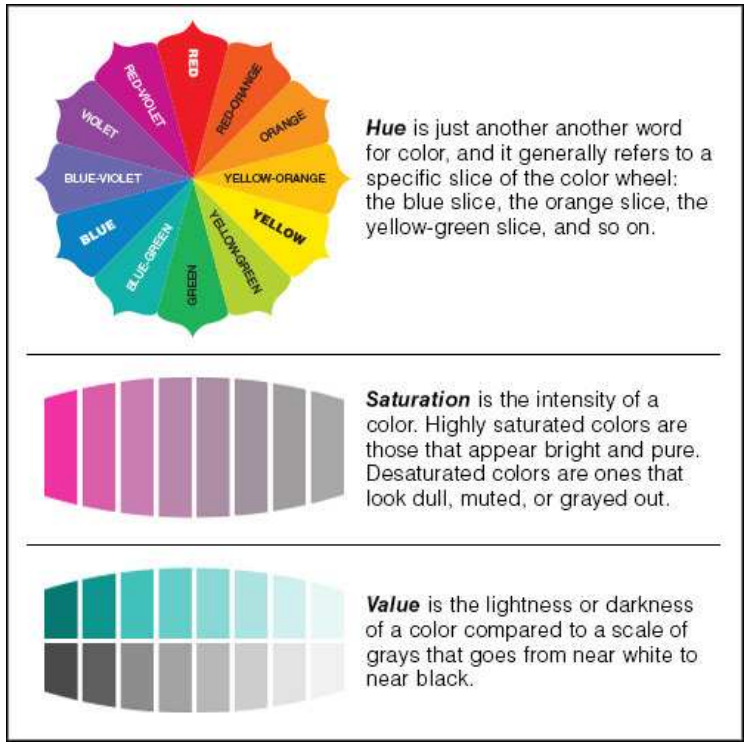


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6.1 Basic Concepts of Images and Graphics



- Color Model : ③ **HSV or HSB model**



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6.1 Basic Concepts of Images and Graphics



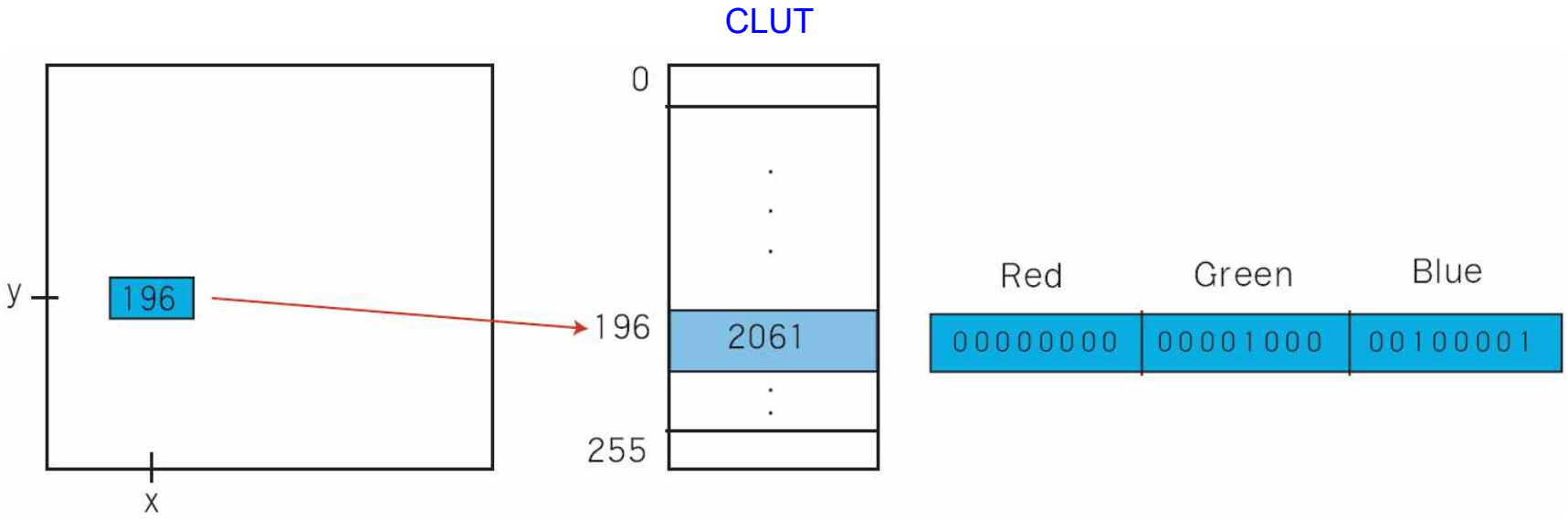
- Color Model : ④ **Indexed Color**
  - Using palette, color look-up table (CLUT)
    - Image representation using predefined colors in a palette (color view table)
    - One point on the screen has a palette number (index), in which the RGB value of the color to be represented is remembered
    - Used on systems with a limited number of available colors
    - Uses a total of 256 colors, usually 8 bits

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6.1 Basic Concepts of Images and Graphics



- Color Model : ④ **Indexed Color**



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6.1 Basic Concepts of Images and Graphics



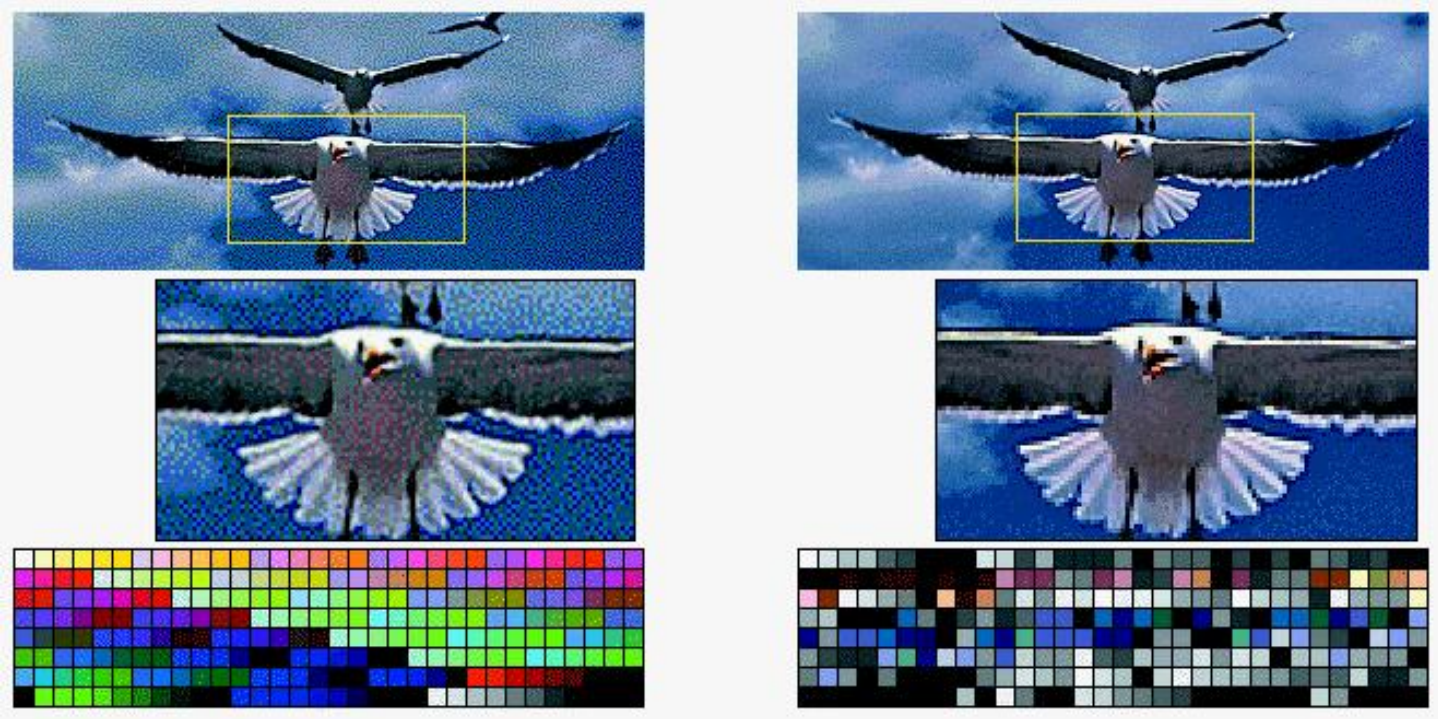
- Color Model : ④ **Indexed Color**
  - System Palettes and Custom Palettes
    - The number of available colors is limited, so color changes can occur
    - Difference in color representation depending on the choice of color suitable for the image

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6.1 Basic Concepts of Images and Graphics



- Color Model : ④ **Indexed Color**



System Palettes and Custom Palettes

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6.2 I / O Device



Image Scanner

- Image input device
  - Convert analog data from documents, photos, films, etc. to digital data
  - The reflected light from the scanned image is received through a CCD (Charge Coupled Device) and the amount of the light is measured
  - Resolution is proportional to the CCD particle's precision, "optical resolution"
  - The scanner consists of a sensor unit, a control unit, an image processing unit, and a data transfer unit.

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6.2 I / O Device



Digital camera

- Save your images using own memory, so pictures directly to computer without printing
- The resolution is expressed as the product of the number of pixels
  - Generally, 2 to 12 million pixels
  - For professional use, it may support more than 15 million pixels
- Most storage methods are JPEG compression
  - Supports uncompressed TIFF or CCD-RAW for higher image quality
- Connect with computer
  - In the past, transmission via serial connection (RS-232)
  - Recently, USB port or Firewire (IEEE1394) port

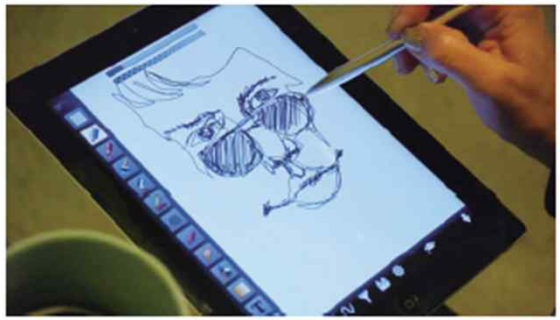
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6.2 I / O Device



Graphic Tablet and Stylus

- Use a pen input device such as a stylus when drawing directly in painting software such as ‘Painter’
- The stylus's pressure forces the tablet to send electrical signals to the computer through the input port
- Coordinates use absolute coordinates on tablet
- More delicate movements and finer operations than the mouse



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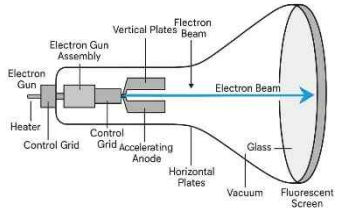
6.2 I / O Device



Display

- Refresh Rate
  - The number of times the screen is displayed per second, in Hertz (Hz)
  - If the activation rate is low, the screen is very flickering
  - Generally, when the frequency is set to 75Hz or higher, you can see the screen without blinking, and VESA recommends 85Hz
- CRT
  - Invented by German scientist Ferdinand Braun in 1897
  - 1940, first use on television
  - Color CRT uses three colors of red, green and blue





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6.2 I / O Device



Display

- LCD (Liquid Crystal Display)
  - The most widely used display in recent years
  - Widely used for monitor, TV, mobile phone screen
- Principle of LCD
  - Grid-shaped cells between two polarizing glass plates, inject liquid crystal
  - Transmits light according to the intensity of voltage
  - Normally, the polarizing filter cuts off the light, passes the voltage when it passes
  - Because it is thin, widely used on wall TV



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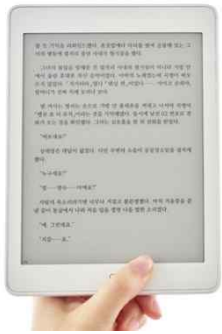
6.2 I / O Device



Display

- OLED(Organic Light-Emitting Diode)
  - Fast response time, high resolution, clear color, low power consumption
  - Expect to replace future monitors and TVs
- Electronic ink (e-ink), electronic paper (e-paper)
  - Display bent like paper
  - Very thin, lightweight and low power consumption
  - Spotlighted as a display of the electronic book dedicated terminal







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6.2 I / O Device



Printer

- Impact method
  - The dot matrix method is typical: the hammer impacts the printer ribbon
  - Noise and print quality are inadequate for multimedia systems



- Non-impact type
  - Laser type: laser to drum, electric field to follow the image to print, print on paper with toner on it
    - Very good print quality but expensive and costly to maintain
  - Inkjet method: jetting fine droplets of ink from a nozzle onto paper
    - Print close to the photo when using high-quality photo paper



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6.3 Processing and compression of images



- Digitizing images
  - Analog image
    - Must be converted to a digital image composed of pixels for processing by a computer
    - Sampling and Quantization Process
  - Sampling
    - When the position data is sampled, the sample point is expressed in pixels
      - That is, the image is divided into pixels
    - The smaller the interval, the higher resolution,
    - The larger the interval, the lower the resolution



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6.3 Processing and compression of images



- Digitizing images
  - Quantization
    - Convert continuous color values to discrete values (quantization levels, pixel values)
      - Sampling process determine sample location, quantization determines the color value at this position
      - The brightness or color of each pixel is expressed as a number (for example, 1 bit, 8 bits, 24 bits)
  - Quantization Error
    - Incorrect edges can occur when the quantization level is insufficient

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6.3 Processing and compression of images



Image Filtering

- Applying any transformation to the image special effects
  - Can restore distorted images to original quality due to noise or distortion
  - Linear and nonlinear filters
  - Real image editing software supports a wide variety of filter techniques

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6.3 Processing and compression of images



Image Filtering

Edge Detection Filter

- Detect the gray level of the image changes suddenly
- As the filter algorithm, we use Sobel algorithm and Kirsch algorithm.





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6.3 Processing and compression of images



Image Filtering

- Average Filter
  - The average value of pixel values around a certain pixel in each pixel of the image
  - Blurred effect as if rubbed by hand
    - Noise Reduction, Edge Blurring
- Brightness Filter
  - Used to change the brightness value of image

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6.3 Processing and compression of images



Image Filtering



Original image



Average filter



Brightness filter

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6.3 Processing and compression of images



Image Filtering

Artistic Filter

- Developed various filters to show artistic effects
- Effects of various brushes and paints such as oil painting or watercolor effect, pencil sketch effect, tearing effect, mosaic effect, knife oil painting



Crystal filter



Watercolor filter

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6.3 Processing and compression of images



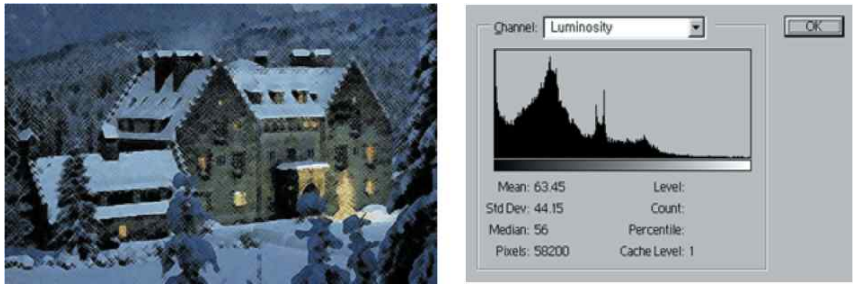
- Image Filtering
  - Histogram Equalization
    - A technique of evenly distributing the number of pixels according to the intensity of an image so that the histogram pushed to one side is widely dispersed as a whole
    - The dark image becomes brighter overall, the brighter image becomes globally darker, and the contrast difference becomes easier to distinguish
    - The bright and dark parts of the image are almost evenly distributed

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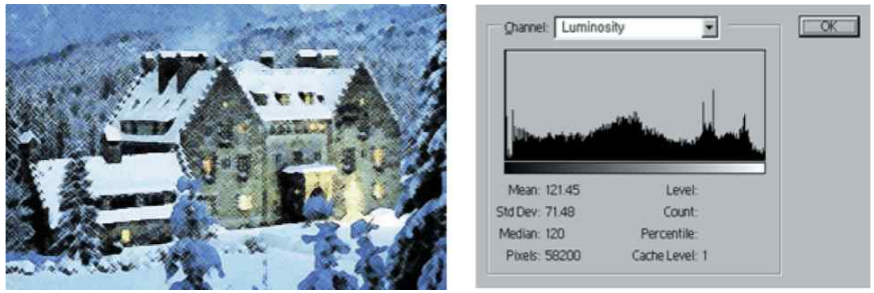
6.3 Processing and compression of images



- Image Filtering
  - Histogram Equalization



Low Contrast



High Contrast

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6.3 Processing and compression of images



- Compress the image
  - The need for data compression
    - saving space
    - faster transmission

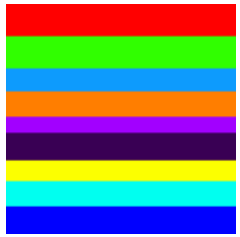
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※ Compression Method

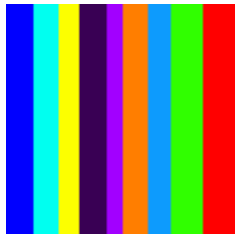


Run Length Encoding

- By indicating how many times the same value repeats,
  - In other words, ABBBBBBBBA is expressed as A1B8A1
- Horizontal compression increases the compression ratio for similar images horizontally
  - For photos, the file size may be rather large
  - Increase compression effect in case of graphic for illustration rather than photo image
  - cf) ABABABABAB



501 bytes



1,148 bytes

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※ Compression Method



Huffman Coding

- Most frequently used characters are converted into small bits of code, and unused characters are converted into many bits of code.
  - It is a way to reduce the amount of bits needed to represent.
- The code is generated and compressed according to the system so as to be compressed as efficiently as possible for each data to be compressed.
  - To do so, a specific code for each character must be determined for each piece of data, but what is needed is a Huffman tree.

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※ Compression Method



- Huffman Coding

Data: Yellow, Red, Red, Red, Blue, Green, Red, Blue, Red, Red

<Code Table>
Red : 00
Blue : 01
Yellow : 10
Green : 11

Code: 10000000011100010000

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※ Compression Method



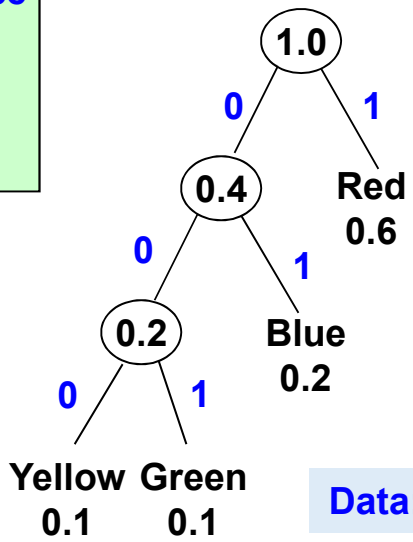
Huffman Coding

Huffman Coding

Freq. of appearance

Red :	0.6
Blue :	0.2
Yellow :	0.1
Green :	0.1

Huffman Tree



Code Table

Red :	1
Blue :	01
Yellow :	000
Green :	001

Result

16bits

0001110100110111

20bits

Code: 10000000011100010000

Data: Yellow, Red, Red, Red, Blue, Green, Red, Blue, Red, Red

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※ Home Work



- Make the Huffman Coding

go go gophers

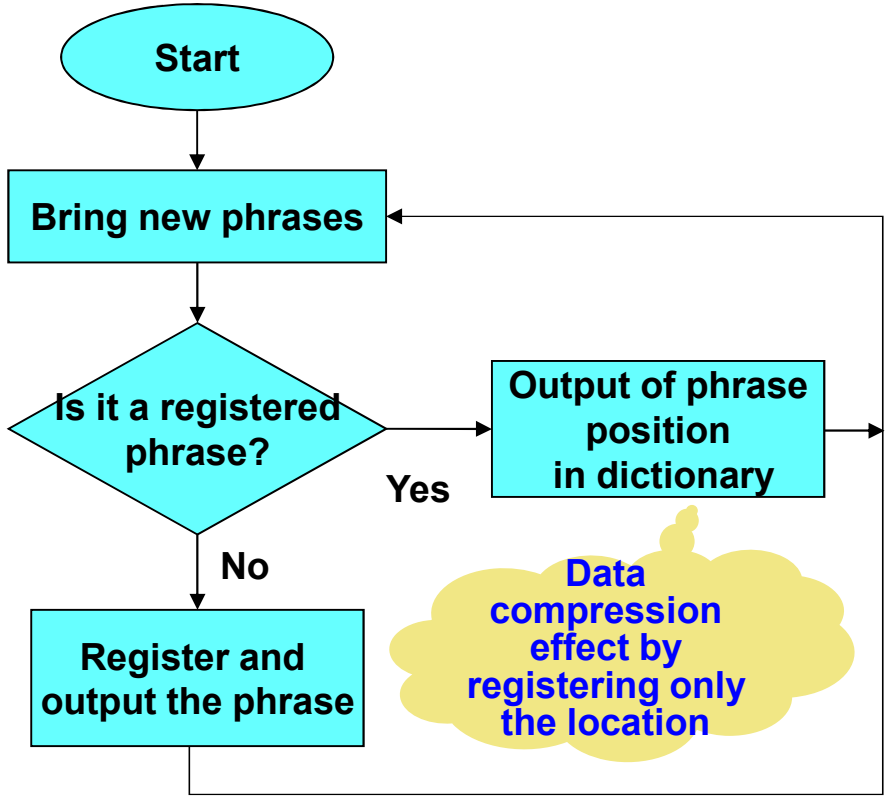
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※ Compression Method



LZ Coding

- Set unit information in text data to one phrase instead of one character
- Several phrases have a repetitive nature.
- If the same phrase appears repeatedly, it is possible to encode the phrases appearing thereafter using the position information on the document where the phrase appears first
- The same phrase can be stored only once



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※ Compression Method



LZ Coding

- We assume a unique dictionary.

D item the 200th word : Dust
C item the 32nd word : cause
S item the 1st word : s
T item the 100th word : the
L item the 150th word : lung
D item the 130th word : disease
P Item the 172nd word : Pneumoconiosis
……..

Dust causes the lung disease Pneumoconiosis, and can cause an allergic reaction in some people
⇨ D 200, C 32, S 1, T 100, L 150, D 130, P172 .....

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※ Compression Method



LZ Coding

- The biggest difficulty of lexical compression method is 'How to construct a dictionary with high compression efficiency by composing a dictionary item'.
- So, a common technique is to search for the most frequently used words by searching frequencies, and assigning priority to long words, among them.

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6.3 Processing and compression of images



- Compress the image
  - Compression method of image data
    - Reduce the amount of data per pixel
      - The smaller the amount of data per pixel, the more difficult it is to express images with subtle density
    - Reduce the number of pixel
      - Difficult to express delicate lines, and mosaic phenomenon occurred
    - Compression method of data
      - Reduce data size while minimizing image degradation
      - Reduce the amount of information in the color space with less visual impact: JPEG compression
      - Compress them when neighboring pixels have the same value: GIF compression

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6.3 Processing and compression of images



GIF(Graphical Interchange Format)

- Image file format developed by CompuServe in 1987
- Popular file formats for transferring raster images from the Internet
- Developed to transfer images quickly through its own online service
- A typical compression format that supports palettes up to 256 colors only
- The size of the file is determined by the number of colors used
- A picture with a small number of colors takes up less memory space than a picture with a large number of colors
- Because colors are only represented with up to 256 colors, it is better to use them for pictures or illustrations than for precise pictures.

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6.3 Processing and compression of images



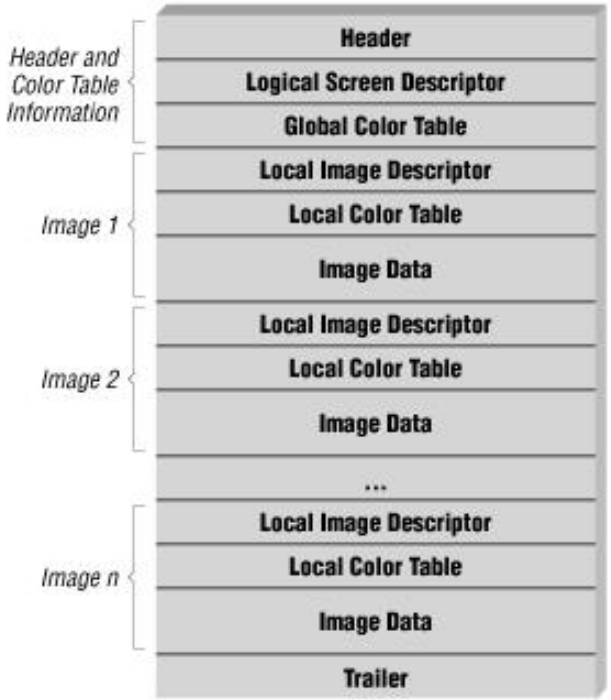
- GIF(Graphical Interchange Format)
  - Basic concept
    - The function of a line or icon made of simple color is used properly
    - Reduce file size by compressing the file size from 1.5: 1 to 2: 1 using the LZW (Lempel-Ziv-Welch) compression method.
    - In the case of photographs, the effect of compression is not significant, but in the case of graphic files produced by illustration,
    - Most widely used with JPEG format on the web
    - There are two types of GIF image formats: GIF89a format and GIF87 format.

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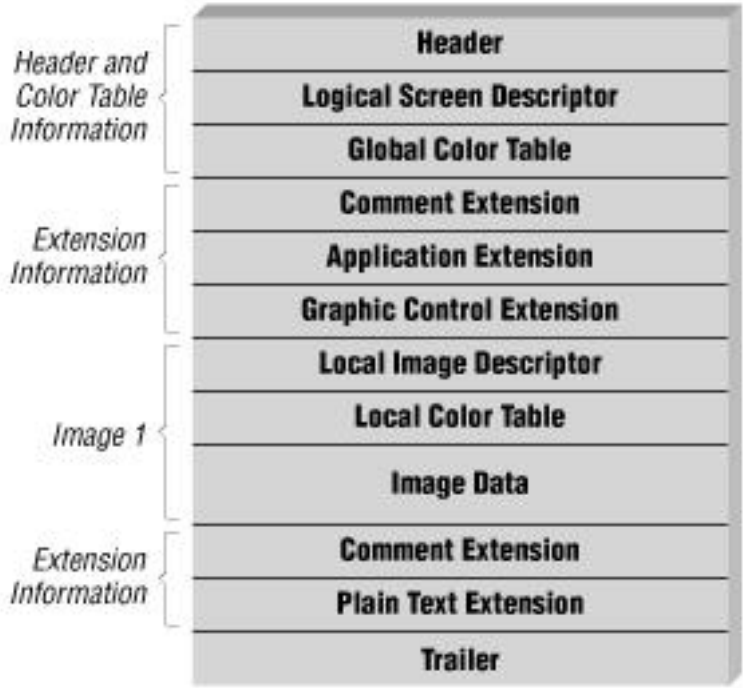
6.3 Processing and compression of images



- GIF(Graphical Interchange Format)



GIF87a file layout



GIF89a file layout

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6.3 Processing and compression of images



- GIF(Graphical Interchange Format)
  - GIF animation





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6.3 Processing and compression of images



- GIF(Graphical Interchange Format)



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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - ISO International Standard
    - 1992, for color still images (pictures)
    - Lossless Compression: When each pixel is important, such as X-rays or CT images
    - Lossy compression: a common method used in JPEG
  - JPEG Features
    - Use 24-bit color
      - Compression characteristics can cause color blurring in the illustration
    - JPEG2000
      - Image Compression Method by Wavelet Technology
      - Includes features such as progressive compression, lossless compression, high compression efficiency and multiple resolutions

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

① Convert from RGB model to YIQ model

- The YIQ model is a model used in NTSC (one of TV broadcasting standards)
- Y includes brightness (or Luminance) information, hue and purity information is included in I and Q
  - Since brightness information is Y, black and white television use only Y value
- Why convert an RGB model to a YIQ model?
  - Focused on the principle that the human vision is more sensitive to the brightness of the image than the color.
  - To more precisely compress Y

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

① Convert from RGB model to YIQ model

- Calculated by simple matrix multiplication below

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

② Macroblock of YIQ

- Split image into 16x16 pixel area of predetermined size
  - 16 × 16 macroblock is the basic unit of compression
- Color Subsampling:
  - Leave the Y macroblock, the I and Q macroblocks are beyond one pixel displayed in 8x8 pixel size
  - Reduce Sampling Size in Less Sensitive Colors (I and Q)

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process
    - ③ Divide the Macro block by 8 × 8 block
      - Compression is performed in units of 8 × 8 pixel blocks
      - The Y, I, Q macroblocks are divided into 4, 1, 1 of 8 × 8 blocks

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

④ DCT (Discrete Cosine Transformation) conversion

- The pixel values of the two-dimensional plane space are converted into two-dimensional frequency information by **Fourier transform**
  - Distinguish between parts that are sensitive to human vision and those that are not.
- Formula for obtaining DCT coefficient matrix B from an N × N pixel block A matrix

$$B(u,v) = \frac{C(u)C(v)}{\sqrt{2N}} \sum_x \sum_y A(x,y) \cos\left\{\frac{(2x+1)u\pi}{2N}\right\} \cos\left\{\frac{(2x+1)u\pi}{2N}\right\}$$

at $C(u) = \begin{cases} \frac{1}{\sqrt{2}} & if\ u = 0 \\ 1 & otherwise \end{cases}$

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

④ DCT (Discrete Cosine Transformation) conversion

- Acquisition of 64 DCT coefficients after DCT conversion
- First value B (0,0): DC (Direct Current) coefficient, average value of 8x8 pixel block
- The remaining 63 : AC (Alternate Current) coefficient, Difference from DC coefficient
  - 64 DCT coefficients are arranged in 8x8 block form
  - DCT coefficients of high frequency are shifted to the lower right position

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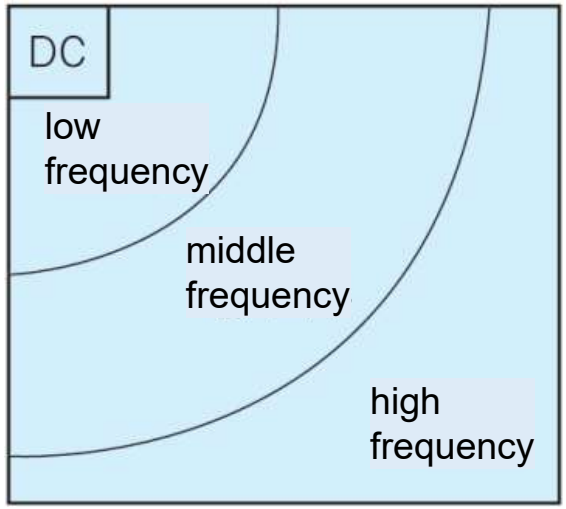
6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

④ DCT (Discrete Cosine Transformation) conversion

- When compressing an image, a high frequency band is a part that has little influence on image recognition



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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

⑤ Quantization

- The process of rounding off DCT coefficients by integers within a range that is difficult to distinguish from human sensory abilities
- Quantization matrix Q: a predefined constant matrix
  - The constant is greater than or equal to 1 and varies in size depending on the frequency
  - The human eye is sensitive to low frequencies, it is divided by a small constant so that there is no big difference from the original value. And high frequencies are not well recognized, it is divided into large numbers

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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

⑤ Quantization

- An example of the elements and Zij matrix Q value for each element of the matrix B of the DCT coefficients in the matrix Z obtained by dividing the element of Q
  - After this process, the coefficients of the high frequency are almost zero.
  - The quantization process is where the largest data compression occurs

$$Z_{ij} = \text{IntegerRound}\{B_{ij} / Q_{ij}\}$$

$$Q = \begin{bmatrix} 3 & 5 & 7 & 9 & . & . & & \\ 5 & 7 & 9 & . & . & & & \\ 7 & 9 & . & . & & & & \\ 9 & . & . & & & & & \\ . & . & & & & & & . \\ . & & & & & & . & 29 \\ & & & & & & 29 & 31 \end{bmatrix}$$

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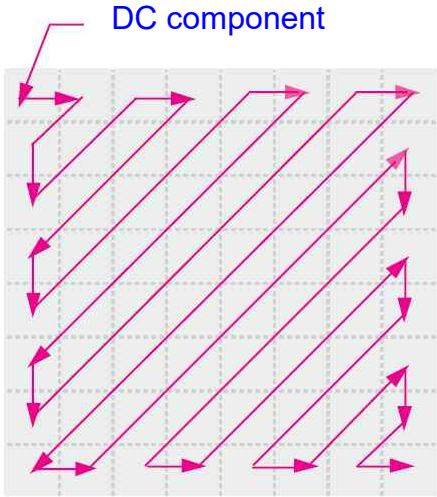
6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

⑥ Zig-zag Scanning

- Since the DC coefficient has a different value from the AC coefficient.
- For easily entropy coding, calculate the DC coefficient and difference of the previous block and store the result
- Then, the coefficients are read in zigzag form and arranged in a one-dimensional form
  - The coefficients of the low frequency are located on the front side and the coefficients of the high frequency are located on the rear side.



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6.3 Processing and compression of images



- JPEG (Joint Photographic Experts Group)
  - JPEG compression process

⑦ Entropy coding

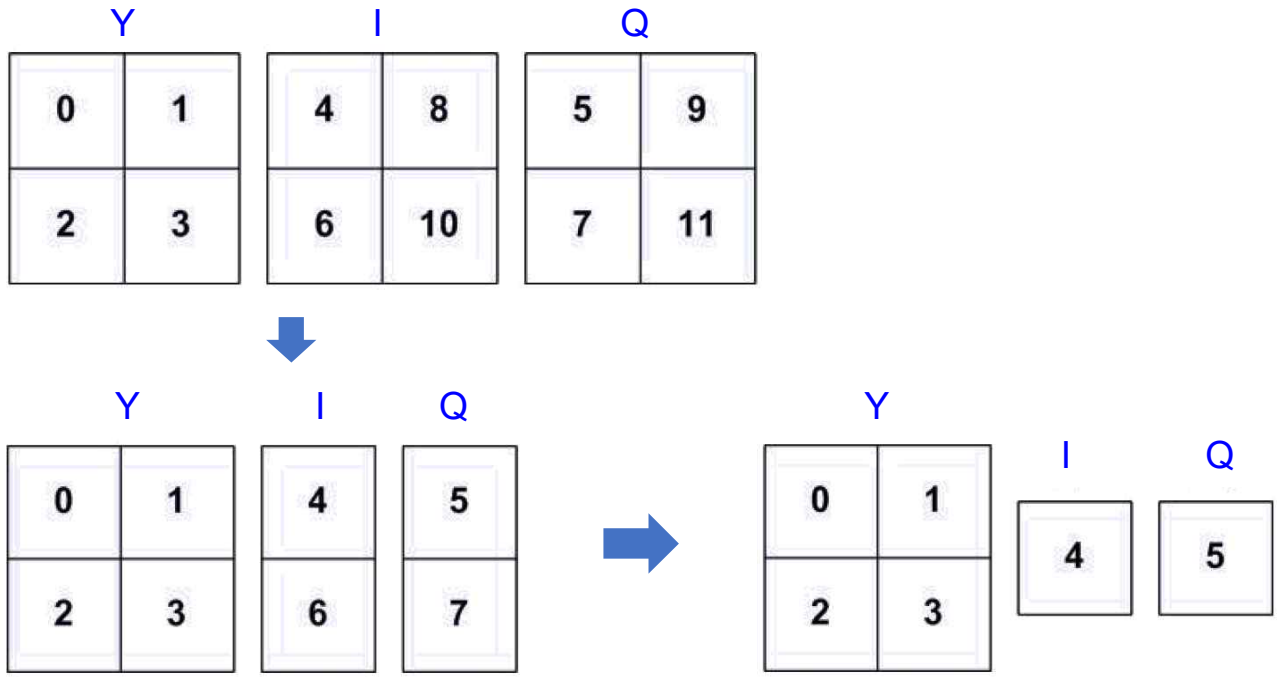
- The process of performing final compression using lossless compression
- Generally, Huffman coding is widely used.

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※ Example of JPEG Encoding



❖ Color Down Sampling



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※ Example of JPEG Encoding



❖ DCT

Image

139	144	149	153	155	155	155	155
144	151	153	156	159	156	156	156
150	155	160	163	158	156	156	156
159	161	162	161	160	159	159	159
159	160	161	162	162	155	155	155
161	161	161	161	161	157	157	157
162	162	161	163	162	157	157	157
162	162	161	161	163	158	158	158

Result of DCT

235.6	−1.0	−12.1	−5.2	2.1	−1.7	−2.7	1.3
−22.6	−17.5	−6.2	−3.2	−2.9	−0.1	0.4	−1.2
−10.9	−9.3	−1.6	1.5	0.2	−0.9	−0.6	−0.1
−7.1	−1.9	0.2	1.5	0.9	−0.1	0.0	0.3
−0.6	−0.8	1.5	1.6	−0.1	−0.7	0.6	1.3
1.8	−0.2	1.6	−0.3	−0.8	1.5	1.0	−1.0
−1.3	−0.4	−0.3	−1.5	−0.5	1.7	1.1	−0.8
−2.6	1.6	−3.8	−1.8	1.9	1.2	−0.6	−0.4

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※ Example of JPEG Encoding



❖ DCT

Result of DCT

235.6	−1.0	−12.1	−5.2	2.1	−1.7	−2.7	1.3
−22.6	−17.5	−6.2	−3.2	−2.9	−0.1	0.4	−1.2
−10.9	−9.3	−1.6	1.5	0.2	−0.9	−0.6	−0.1
−7.1	−1.9	0.2	1.5	0.9	−0.1	0.0	0.3
−0.6	−0.8	1.5	1.6	−0.1	−0.7	0.6	1.3
1.8	−0.2	1.6	−0.3	−0.8	1.5	1.0	−1.0
−1.3	−0.4	−0.3	−1.5	−0.5	1.7	1.1	−0.8
−2.6	1.6	−3.8	−1.8	1.9	1.2	−0.6	−0.4

÷

Quantization matrix :
control the compression ratio
(for a quality of 50% as
specified in the original JPEG
Standard)

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

=

Result of Quantization

15	0	−1	0	0	0	0	0
−2	−1	0	0	0	0	0	0
−1	−1	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

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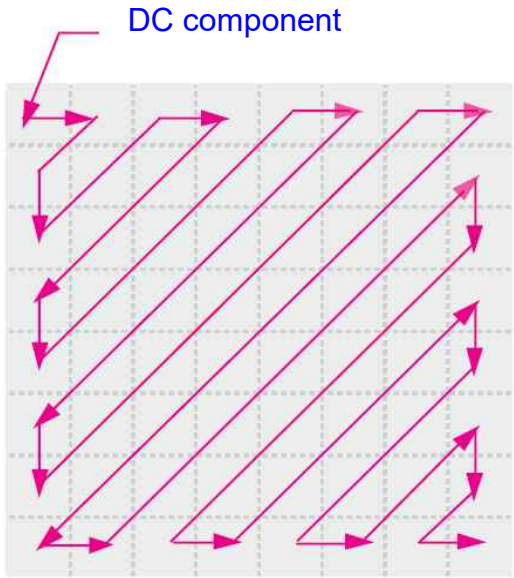
※ Example of JPEG Encoding



❖ DCT

Result of Quantization

15	0	−1	0	0	0	0	0
−2	−1	0	0	0	0	0	0
−1	−1	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0



Zig-zag Scanning

[15, 0, -2, -1, -1, -1, 0, 0,-1, …, 0]

⇩

[15, 0, -2, -1, -1, -1, 0, 0,-1, EOB]

Entropy coding

[01100110110100000000011100010 10] ⇨ **31 bits**

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※ Example of JPEG Encoding



- Performance of compression
  - Express the pixel value with 8 bits: The total data is 64 × 8 = 512 bits
  - Perform JPEG compression: reduced to 31 bits

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※ Example of JPEG Encoding



❖ Decoding

240	0	−10	0	0	0	0	0
−24	−12	0	0	0	0	0	0
−14	−13	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

IDCT

144	146	149	152	154	156	156	156
155	156	152	154	156	156	156	156
155	156	157	158	158	157	156	155
160	161	161	162	161	159	157	155
163	163	164	163	162	160	158	156
163	164	164	164	162	160	158	157
160	161	162	162	162	161	159	158
158	159	161	161	162	161	159	158

Original Image

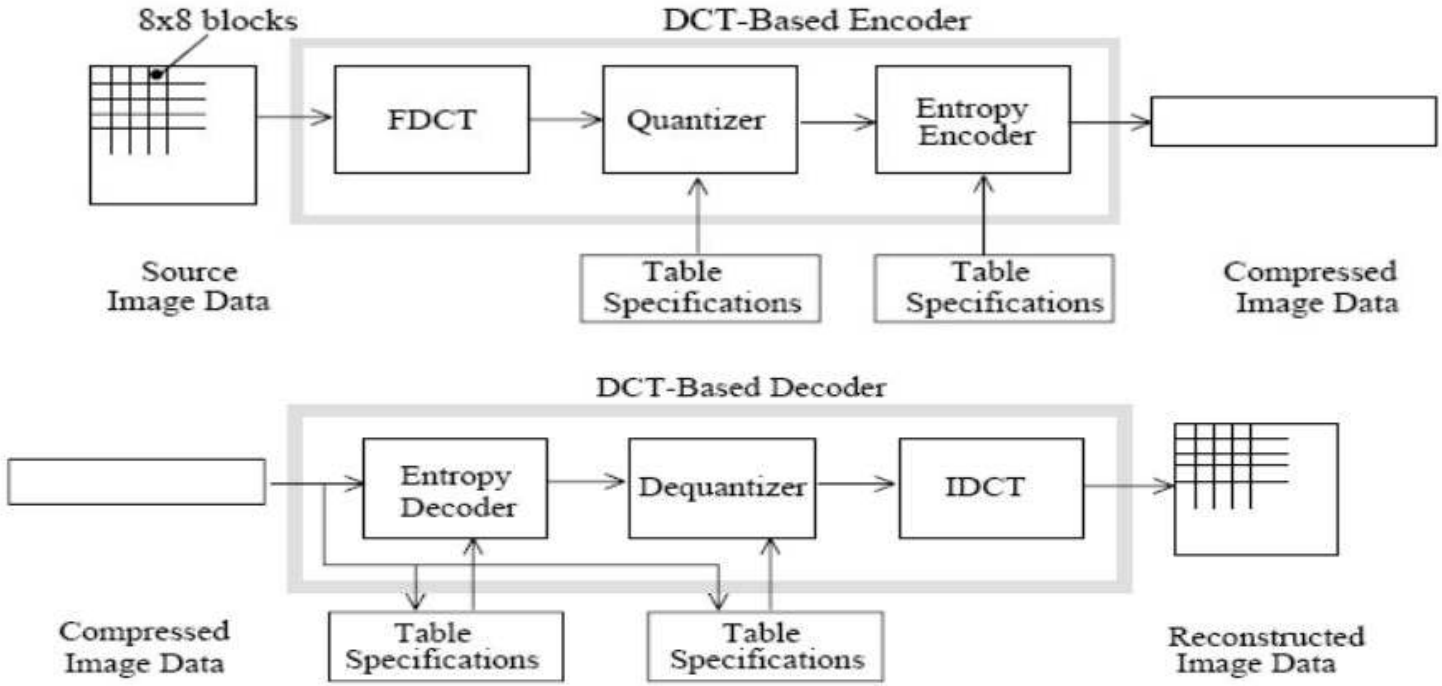
139	144	149	153	155	155	155	155
144	151	153	156	159	156	156	156
150	155	160	163	158	156	156	156
159	161	162	161	160	159	159	159
159	160	161	162	162	155	155	155
161	161	161	161	161	157	157	157
162	162	161	163	162	157	157	157
162	162	161	161	163	158	158	158

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6.3 Processing and compression of images



- Compression and decompression process of JPEG



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6.3 Processing and compression of images



JPEG2000

- In the case of JPEG, wavelet technology is better than DCT, which is the basic method
- Development of still image compression standard called wavelet-based JPEG2000



Original image 3Mbyte



JPEG 19Kbyte



JPEG2000 19Kbyte

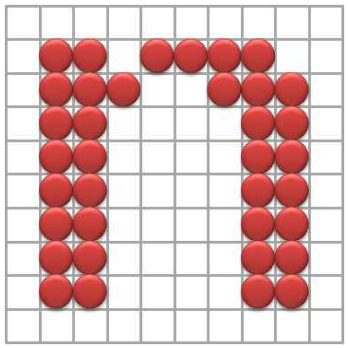
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6.4 2D / 3D Graphics



Character Font

- Raster Font (Bitmap Font)
  - Remember the location of the pixels to represent the letters ⇒ Bitmap
  - Advantages: fast display on screen
  - Disadvantages: Staircase phenomenon appears on enlargement
  - Used for low-resolution printer / display devices



I am RASTER Font

I am RASTER Font

I am RASTER

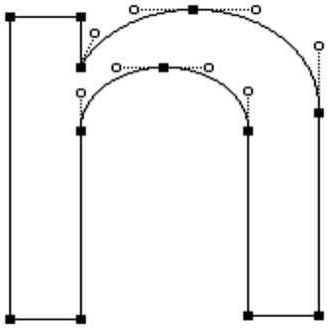
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6.4 2D / 3D Graphics



Character Font

- Vector Font
  - Remember the line type / coordinates and their arguments
  - Pros: Keeps clean text even when zoomed in
  - Disadvantages: it takes more time to display because it requires calculation
  - Currently Windows and printers are mostly vector fonts



I am VECTOR Font

I am VECTOR Font

I am VECTOR

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6.4 2D / 3D Graphics



- The primitives composing the 2D vector graphics include points, lines, circles, ellipses, polygons, curves, etc.
  - Line drawing
  - Filling of Region and Polygon
  - Antialiasing

→ Ref) Computer Graphics

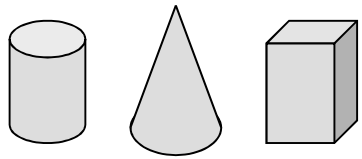
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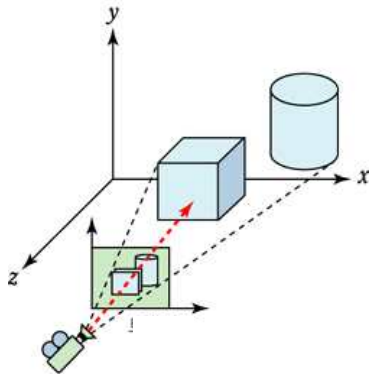
6.4 2D / 3D Graphics

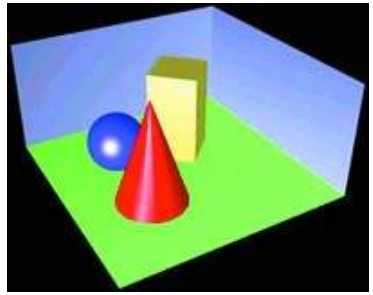


3D graphics

- The main purpose of 3D graphics is real feeling effect
  - Three-dimensional representation of objects that do not exist in the real world
- 3D graphic creation process
  - Modeling the geometric shape of an object
  - Project a three-dimensional object onto a two-dimensional plane (Projection)
  - Drawing color and contrast of generated 3-D objects (Rendering)





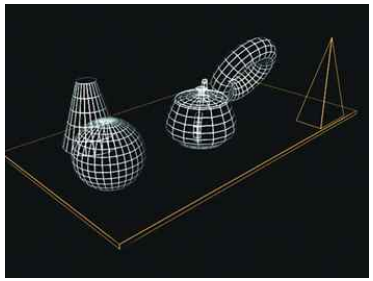


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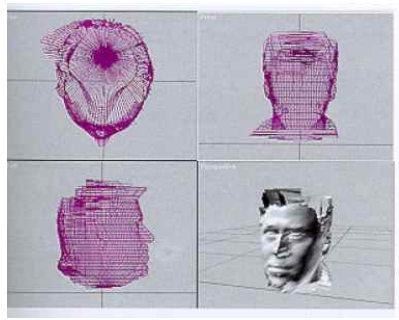
6.4 2D / 3D Graphics



- 3D graphics
  - Modeling
    - The process of expressing a shape with a three-dimensional coordinate system
      - Wireframe model
      - Polygon Surface Model
      - Solid model
    - Modeling by 3D scanning
      - Scanning real human faces or objects
      - 3D digitizer, 3D laser scanner





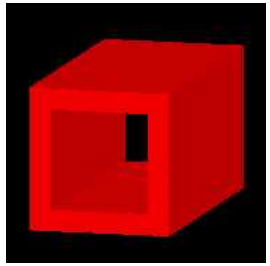


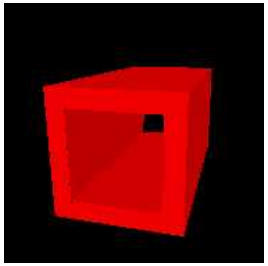
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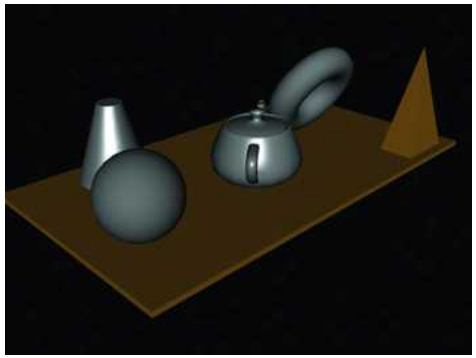
6.4 2D / 3D Graphics



- 3D graphics
  - Projection
    - Project a three-dimensional object onto a two-dimensional plane
    - Parallel projection and perspective projection
  - Rendering
    - The process of adding realism by adding a three-dimensional texture such as a change of shadows or colors
    - Used to transform a wireframe image into a shaded image
    - Includes Hidden Surface Removal, Shading, and Texture Mapping





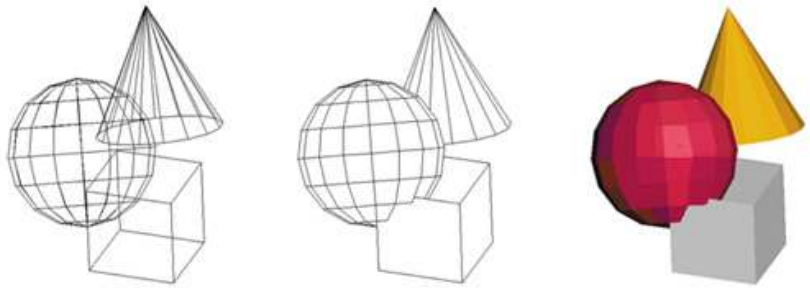


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6.4 2D / 3D Graphics



- 3D graphics
  - Hidden Surface Removal
    - Z-buffer technique is often used
    - Determine whether each pixel can see an object from the camera's viewpoint
    - Compares depth values of objects and draws only the closest object
  - Shading
    - For the three-dimensional effect of an object, color and contrast are expressed on the surface of the object.
    - Ray Tracing Technique



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6.4 2D / 3D Graphics



- 3D graphics
  - Texture Mapping
    - Not only two-dimensional texture such as color or brightness but also three-dimensional characteristics such as transparency and reflection of an object
    - The texture is defined, and the surface of the 3D object is drawn through the texture mapping process



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6.5 File formats for images and graphics



- Raster file format
  - BMP
    - Supported by Microsoft, the most basic format in bitmap
    - Disadvantages of large files because they are not compressed
  - Tagged Image File Format (TIFF)
    - Developed jointly by Aldus and Microsoft
    - Developed as a common file format for use in scanners in the 1980s
    - Used in tabletop publishing, fax, medical image processing and 3D applications
  - GIF
    - Graphic files for illustrations have a high compression effect
    - GIF89, GIF89a: Transparent color and animation

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6.5 File formats for images and graphics



- Raster file format
  - Portable Network Graphics (PNG)
    - Lossless graphics file format
    - Developed in open format to improve GIF patent issues and 256 color limit
    - In most cases, PNG is more compressible than GIF and supports 24-bit true color, allowing you to store all of your original colors
    - Animation is not supported
  - Joint Photographic Experts Group (JPEG)
    - File formats specially designed for photo compression
    - Utilizing the fact that human eyes perceive contrast better than color
    - Lossy Compression
    - JPEG is suitable for handling large images with many colors

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6.5 File formats for images and graphics



Comparing File Formats

Original BMP	JPEG	GIF	Original BMP	JPEG	GIF
					
416KB	19.8KB(21.0:1)	4.91KB(84.7:1)	1641KB	84.2KB (19.49:1)	168.0KB(9.77:1)

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6.5 File formats for images and graphics



- Vector graphics file format
  - Encapsulated Postscript (EPS)
    - Utilizes the Postscript language used in printers
    - Display graphic structure, font, bitmap information of text
  - Windows Metafile (WMF)
    - Metafile format used by Windows, used for office clip art
    - It is most suitable if you want to express both bitmap and vector information together.
  - AI
    - File formats used in Adobe Illustrator
  - CDR
    - File formats used in Corel Draw

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6.5 File formats for images and graphics



- 3D graphic file format
  - 3DS
    - File formats used in 3D Studio
    - Used in most 3D graphics software
  - DXF
    - Developed for use on its AutoCAD from Autodesk Inc.
    - Widely used in CAD software
  - WRL
    - Format developed for VRML
    - Configuration, and position information of the object on the 3D object

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※ Home Work



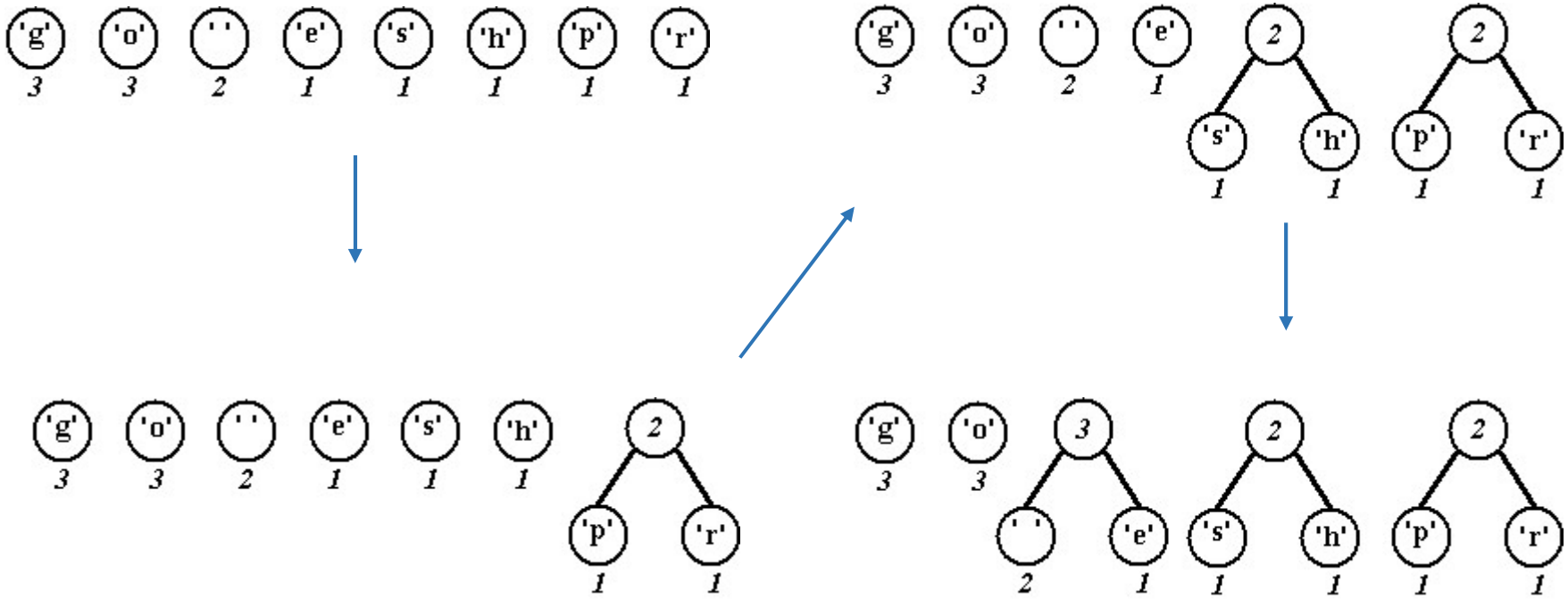
- Make the Huffman Coding

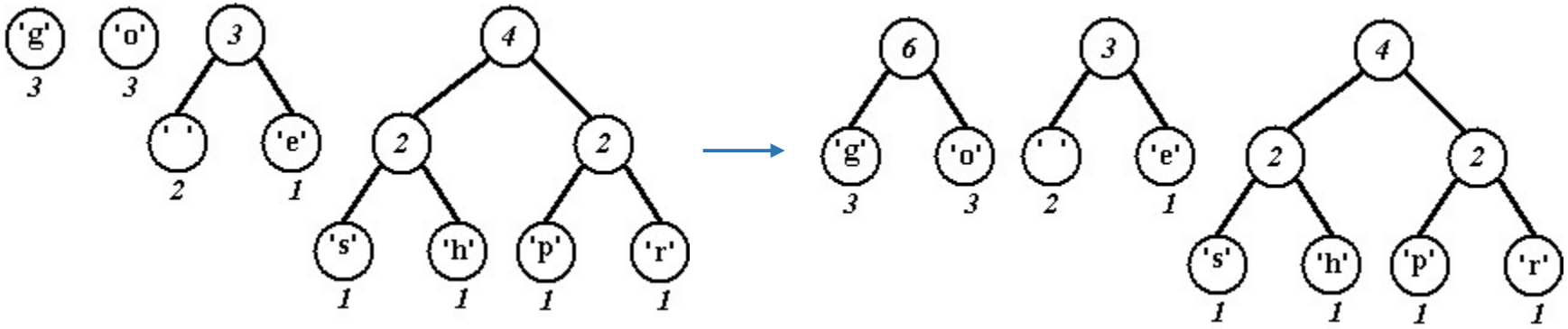
go go gophers

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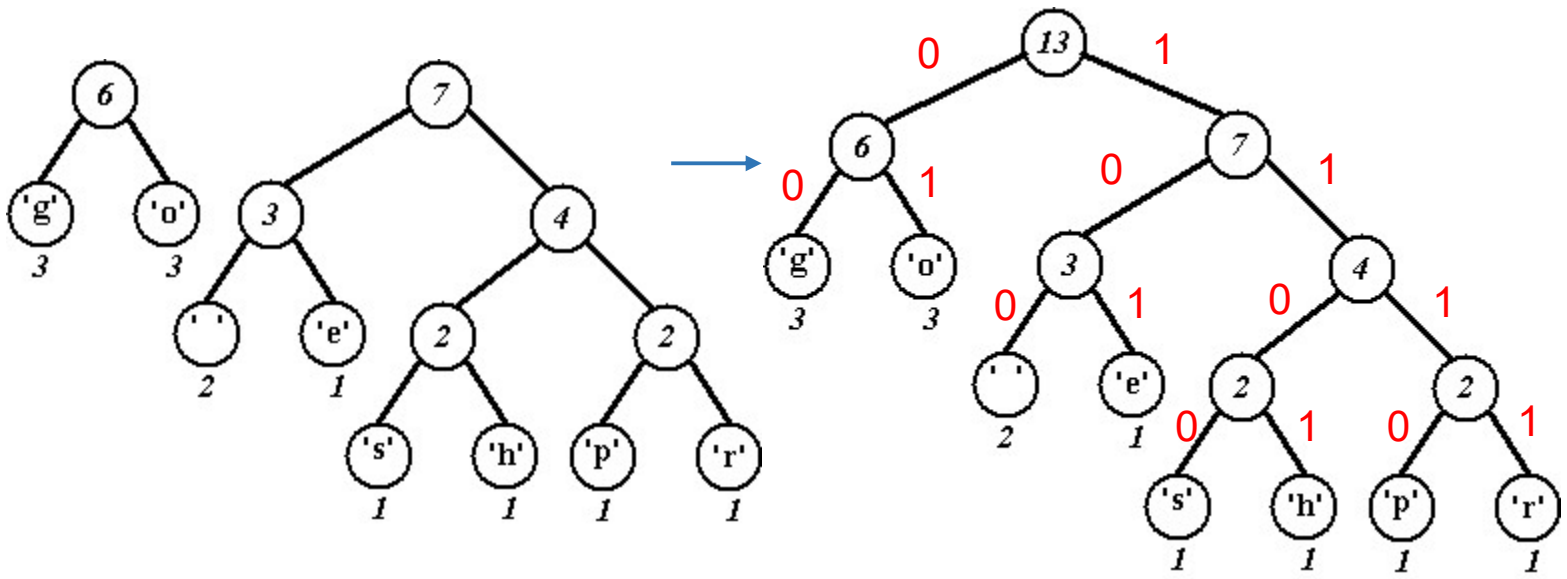


go go gophers





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char	binary
'g'	00
'o'	01
'p'	1110
'h'	1101
'e'	101
'r'	1111
's'	1100
' '	100

00	01	100	00	01	100	00	01	1110	1101	101	1111	1100
g	o		g	o		g	o	p	h	e	r	s

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Summary

Basic Concepts of Images and Graphics
- Pixel, Resolution
- Color Model: RGB, CMY, HSV

Image Processing
- Image Device
- Compression

Graphic File Format
- JPEG, GIF, BMP
- Fonts

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